

1	Course Name:
	Programming for Data Analytic
2	Course Code:
	IIB40303
3	Year and Semester Offered:
	Year 2 Semester 2
4	Credit Hours Value:
	3 Credit Hours [3 SLT Credit]
5	Prerequisite(s), if applicable:
	IGB20303 Probability and Statistics for IT
6	Course Synopsis:
	The course discusses on how best to make sense of voluminous data and potentially fast-changing information, especially when data is used to take real-time decisions (about investments, policies, trades and new markets etc.) or it needs to be presented to wider audiences. The use of data analytics is increasing in the business for better decision making and utilizing the pile of data collected through many business procedures (purchase, selling, supply chain etc.). Consequently, we are experiencing a period of rapid innovation in new data science techniques intended to serve this purpose. In this course, we examine the data analytics aspects provided in python and the emerging professional practices of turning raw data into meaningful knowledge.
7	Course Learning Outcome:
	Upon completion of the course, students should be able to: 1. Understanding the programming procedures required to conduct data analytic. 2. Demonstrate strong programming and analytical skills to solve broadly defined in IoT. 3. Design programming solutions for broadly-defined IT and IoT related problems. 4. Demonstrate programming knowledge apply procedures, processes, systems and methodologies.
8	Content Outline:
	Topic
	1. Descriptive Statistics
	2. Data Gathering and Cleaning
	3. Bivariate Statistics
	4. Introduction to Regression
	5. Machine Learning: Introduction and Concepts
	6. Case Study